Electromagnetism, Physical Optics and Photonics

Science

May, 2026

Ollscoil Teicneolaíochta an Oirdheiscirt
South East Technological University

Electromagnetism, Physical Optics and Photonics (A11478)

Short Title:	Elect, Phys Optics & Photonics
Faculty	Science and Computing
Department	Science
Cluster	Mathematics and Physics
Author	JHOULIHAN
Credits	5

Level: Intermediate

Description of Module / Aims

This module develops concepts in optics from electromagnetic theory.

Programmes

ELEC-0040	BSc (Hons) in Physics for Modern Technology (WD_KPHTE_B)	3 / 5 / M

Indicative Content

- Harmonic waves; phase and phase velocity
- Complex representation
- Plane and spherical waves
- Maxwell's equations and electromagnetic waves; energy and momentum of electromagnetic waves, Poynting vector
- Polarisation of electromagnetic waves
- Superposition principle; complex methods and phasor addition
- Phase and group velocities
- Fourier analysis; periodic and non-periodic waves
- Pulses and wave packets; optical bandwidth
- Mathematical representation of polarised light
- Dichroism, polarisation by scattering and reflection
- Birefringence, double refraction and optical activity
- Fraunhofer and Fresnel diffraction
- The grating equation, free spectral range, dispersion, resolution; blazed gratings; interference gratings
- Introduction to lasers: elements of laser, gain-medium and resonant cavity; elementary description of 3- and 4-level lasers; HeNe and semiconductor lasers; applications in science, engineering, medicine

Learning Outcomes

On successful completion of this module, a student will be able to:

1. Analyse the theory underpinning the mathematical representation of electromagnetic waves and physical optics.
2. Apply electromagnetic theory to the aspects of polarisation, interference, diffraction, scattering, and to waveguide propagation.
3. Analyse the theory underpinning the operation of key photonic devices and a range of their applications in telecommunications, medical science, metrology.
4. Demonstrate practical experience in the investigation and analysis of complex optical phenomena.

Learning and Teaching Methods

- Lectures.

- Practical sessions.
- Problem-solving sessions.

Assessment Methods

	Weighing	Outcomes Assessed
Final Written Examination	70%	1,2,3
Continuous Assessment	30%	
Practical	30%	4

Assessment Criteria

<40%: Inability to understand the basic concepts encountered in this module.

40%-49%: Ability to demonstrate a basic understanding of the mathematical representation of electromagnetic waves and related phenomena.

50%-59%: Ability to understand and explain advanced concepts related to electromagnetic waves.

60%-69%: All the above, and in addition, apply problem-solving techniques to complex situations involving electromagnetic waves.

70%-100%: All previous to an excellent level. Demonstrates an ability to comprehensively discuss the course material and analyse novel electromagnetic wave situations.

Learning Modes

Learning Mode	F/T Hours	P/T Hours
Lecture	24	
Tutorial	6	
Practical	18	
Independent Learning	87	

Essential Material

- Graham Smith, F. and T.A. King. _Optics and Photonics_. New York: Wiley, 2002.
- Hecht, E. _Optics_. New York: Addison Wesley Longman, 2000.
- Pedrotti, F.L. and L.S. Pedrotti. _Introduction to Optics_. London: Prentice-Hall, 2002.
- Wilson, J. and J. Hawkes. _Optoelectronics_. London: Prentice-Hall, 1998.

Requested Resources

- SCIENCE LAB: Physics